

# OSCAR DAVIS

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## EDUCATION

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- PhD in Computer Science**, University of Oxford Oct 2023 – Dec 2026  
• Funded by Project CETI and Intel. Supervised by Prof M. Bronstein, Dr İ. Ceylan, Dr J. Bose.
- MSc in Advanced Computer Science**, University of Oxford Oct 2022 – Aug 2023  
• Supervised by Prof M. Bronstein and Dr İ. Ceylan. Obtained with distinction. Best dissertation prize (see below).
- Visting Student**, Imperial College, London Sep 2021 – Jul 2022  
• Supervised by Prof A. Gervais. Funded by Swiss scholarship. Finished with distinction.
- BSc in Computer Science**, EPFL Sep 2019 – Jul 2022

## RESEARCH EXPERIENCE

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- Research Intern at Apple MLR**, with Prof M. Cuturi Mar 2026 – Present
- Research Intern at Genesis Molecular AI**, with Dr J. Bose, Dr N. Boffi, Dr M. Al-Shedivat Jan – Mar 2026  
• Flow maps, steering and tilting for protein generation.
- Research Intern at Microsoft Research, Cambridge**, with Dr J. Gladrow & Dr K. Kalinin Nov 2023 – Feb 2024  
• Engineering work on Diffusion Models, Latent Diffusion Models, VAEs, simple video models, Neural ODEs.  
• Theoretical analyses of Diffusion Models via SDEs, PDEs. (*Patented.*)
- MSc Dissertation**, Information Theory for GNNs, with Dr. İ. Ceylan, Prof. M. Bronstein Feb – Aug 2023  
• Developed a formal information-theoretic framework to fully characterise informational bottlenecks in Graph Neural Networks, including over-smoothing and over-squashing. **Received the Tony Hoare Prize for the best dissertation of the year.**
- BSc Research Project**, DeFi analysis, with Dr A. Gervais Jan – Aug 2022  
• Analysed DeFi markets on the Ethereum and BNB Chain blockchains, quantified offered financial security.  
• Created a program in Go using a custom GPU version of Bellman-Ford in CUDA to detect real-time arbitrage opportunities, and to quantify historically how much more assets could have been extracted, scanning  $864 \times$  more markets than previous SOTA within  $1.5 \pm 1.2$  seconds, outperforming past arbitrage by on average 0.06 ETH and up to 4.4 ETH.
- Student Research Project**, Scala 3.0 Compiler Extension, with Prof M. Odersky Jun – Sep 2021  
• Started and contributed to the thread-safe re-implementation of “lazy-vals”, in the Scala 3.0 compiler.

## PUBLICATIONS

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- Scaling Categorical Flow Maps** May 2026  
[Davis, O.](#), Filippova, A., Ablin, P., Turrise, V., Shidani, A., Cuturi, M., Béthune, L.  
Pre-print, arXiv: [arxiv.org/abs/2605.07820](https://arxiv.org/abs/2605.07820).
- Categorical Flow Maps** Jan 2026  
Roos, D.\*, [Davis, O.\\*](#), Eijkelboom, F.\*, Bronstein, M., Welling, M., Ceylan, İ., Ambrogioni, L., vd Meent, J.W.  
ICML 2026. arXiv: [arxiv.org/abs/2602.12233](https://arxiv.org/abs/2602.12233). GitHub: [github.com/olsdavis/semicat](https://github.com/olsdavis/semicat).
- Generalised Flow Maps for Few-Step Generative Modelling on Riemannian Manifolds** Sep 2025  
[Davis, O.](#), Boffi, N., Albergo, M., Bronstein, M., Bose, J.  
ICLR 2026. NeurIPS 2025 FPI. arXiv: [arxiv.org/abs/2510.21608](https://arxiv.org/abs/2510.21608). GitHub: [github.com/olsdavis/gfm](https://github.com/olsdavis/gfm).

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| <b>SOAPIA: Siamese-Guided Generation of Off Target-Avoiding Protein Interactions [...]</b><br>Vincoff, S.*, <b>Davis, O.*</b> , Tong, A., Bose, J., Chatterjee, P.<br>ICML 2025 FM4LS. OpenReview: <a href="https://openreview.net/pdf?id=Ax25SLIDsN">openreview.net/pdf?id=Ax25SLIDsN</a> .                                                                    | May 2025 |
| <b>FORT: Forward-Only Regression Training of Normalizing Flows</b><br>Rehman, D., <b>Davis, O.</b> , Lu, J., Tang, J., Bronstein, M., Bengio, Y., Tong, A., Bose, J.<br>ICLR 2026. ICML 2025 GenBio ( <b>Best paper award</b> ). arXiv: <a href="https://arxiv.org/abs/2506.01158">arxiv.org/abs/2506.01158</a>                                                 | May 2025 |
| <b>SOAPI: Siamese-guided Generation of Off-Target-Avoiding Protein Interactions</b><br>Vincoff, S., <b>Davis, O.</b> , Tong, A., Bose, J., Chatterjee, P.<br>ICLR 2025 GEM ( <b>Spotlight</b> ). OpenReview: <a href="https://openreview.net/pdf?id=aRrXs2cVdy">openreview.net/pdf?id=aRrXs2cVdy</a> .                                                          | Mar 2025 |
| <b>Fisher Flow Matching for Generative Modeling over Discrete Data</b><br><b>Davis, O.</b> , Kessler, S., Petrache, M., Ceylan, I., Bronstein, M., Bose, J.<br>NeurIPS 2024. arXiv: <a href="https://arxiv.org/abs/2405.14664">arxiv.org/abs/2405.14664</a> . GitHub: <a href="https://github.com/olstdavis/fisher-flow">github.com/olstdavis/fisher-flow</a> . | May 2024 |

## TEACHING EXPERIENCE

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| <b>Co-Lead TA for generative modelling at EEML 2025</b> , Sarajevo (Bosnia and Herz.)<br>• Writing and presenting a geometric generative modelling tutorial (flow matching, Riemannian flow matching).   | Jul 2025                |
| <b>Graduate Teaching and Research Scholarship in CS</b> , Oriel College, Oxford<br>• Teaching undergraduate-level courses to students of Oriel College. Admissions interviews.                           | Apr 2025 – Mar 2026     |
| <b>TA for Geometric Deep Learning</b> , University of Oxford, under Prof. M. Bronstein<br>• Teaching PyTorch implementations of geometric models (equi-/invariance) and others (e.g., neural diffusion). | Jan – Mar 2025 and 2026 |
| <b>OÄW Winter AI School 2025</b> , OÄW, Vienna (Austria)<br>• Gave two PyTorch tutorials: one on implementing Graph Neural Networks; one on (Riemannian) flow matching.                                  | Jan 2025                |
| <b>TA for Graph Representation Learning</b> , University of Oxford, under Dr. I Ceylan<br>• Teaching PyTorch and PyTorch Geometric (for Graph Neural Networks, and Knowledge Graph Learning).            | Oct – Dec 2023 and 2024 |
| <b>TA for Object-Oriented Programming (Java)</b> , EPFL, under Dr M. Schinz<br>• Second most prolific helper on the student forum. Leader of marking group for final projects.                           | Feb – Jun 2020          |

## ACADEMIC ACHIEVEMENTS & OTHERS

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| <b>ICML 2025 GenBio – Best paper award</b><br>• For <i>FORT: Forward-Only Regression Training of Normalizing Flows</i> .                                                                  | May 2025 |
| <b>G-Research Grant for PhD Students and Postdocs</b> (£1k)                                                                                                                               | Feb 2024 |
| <b>Tony Hoare Prize for the best MSc Dissertation</b> , University of Oxford<br>• Prize awarded for my dissertation titled “Information-Theoretic Perspectives on Graph Neural Networks.” | Sep 2023 |
| <b>Swiss Study Foundation Scholarship</b><br>• Granted based on academic performance (almost 100% GPA on my last term’s exams).                                                           | Sep 2021 |
| <b>French Scientific Baccalaureate with Advanced Mathematics</b><br>• Obtained high honours, and 100% in Mathematics, with the Advanced Mathematics option.                               | 2019     |
| <b>French National Mathematics Olympiads</b><br>• Obtained a distinction in the Bordeaux academy.                                                                                         | 2017     |

## SERVICE

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| <b>NeurIPS Top Reviewer</b> | 2025 |
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